

FireRisk AI: Weather-Based Forest Fire Occurrence Prediction

Statistics Mini Project | Python | Supervised Learning

Abstract

This mini project investigates whether meteorological conditions and Fire Weather Index (FWI) variables can be used to classify daily observations into fire and not-fire categories. The study uses 122 observations from the Bejaia region of the Algerian Forest Fires dataset. The workflow includes data preprocessing, descriptive statistics, exploratory data analysis, statistical significance testing, Logistic Regression and Random Forest classification, and model evaluation.

1. Problem Statement

Forest fires are affected by environmental and fire-weather conditions. The objective is to determine whether a supervised machine-learning model can classify a daily observation as fire or not fire using measured weather variables and Fire Weather Index components.

2. Dataset Description

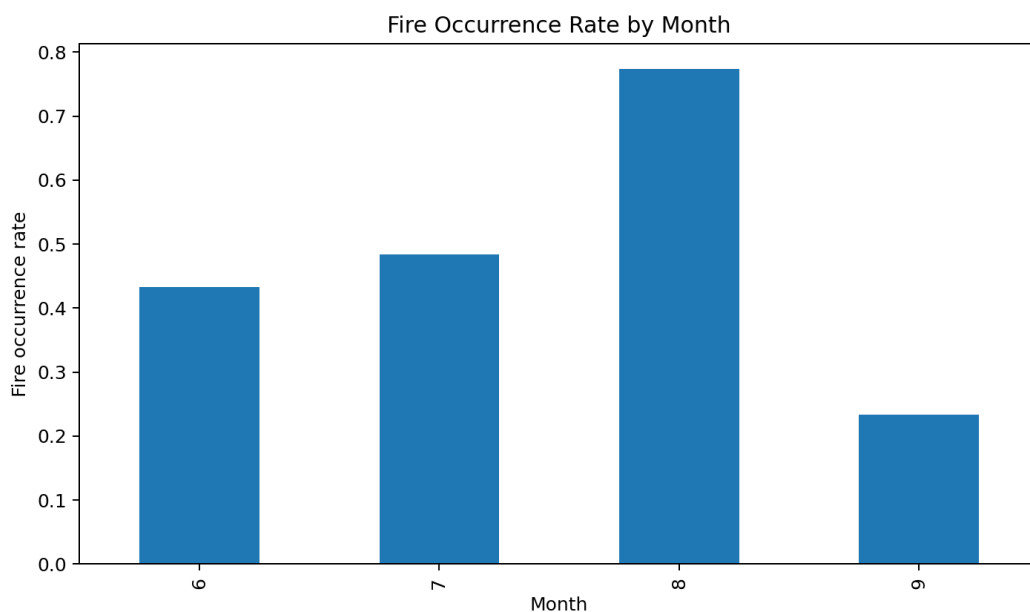
The original UCI Algerian Forest Fires dataset contains 244 observations from two Algerian regions during June–September 2012. This project uses the 122 observations from the Bejaia region. The target is Classes: fire or not fire.

3. Data Preprocessing

The class labels were cleaned and encoded as 1 for fire and 0 for not fire. Data types and missing values were checked. No missing values were found in the working dataset. Standardization was applied within the Logistic Regression pipeline.

4. EDA

Fire observations had higher average temperature and Fire Weather Index variables, while not-fire observations had higher relative humidity and rainfall.



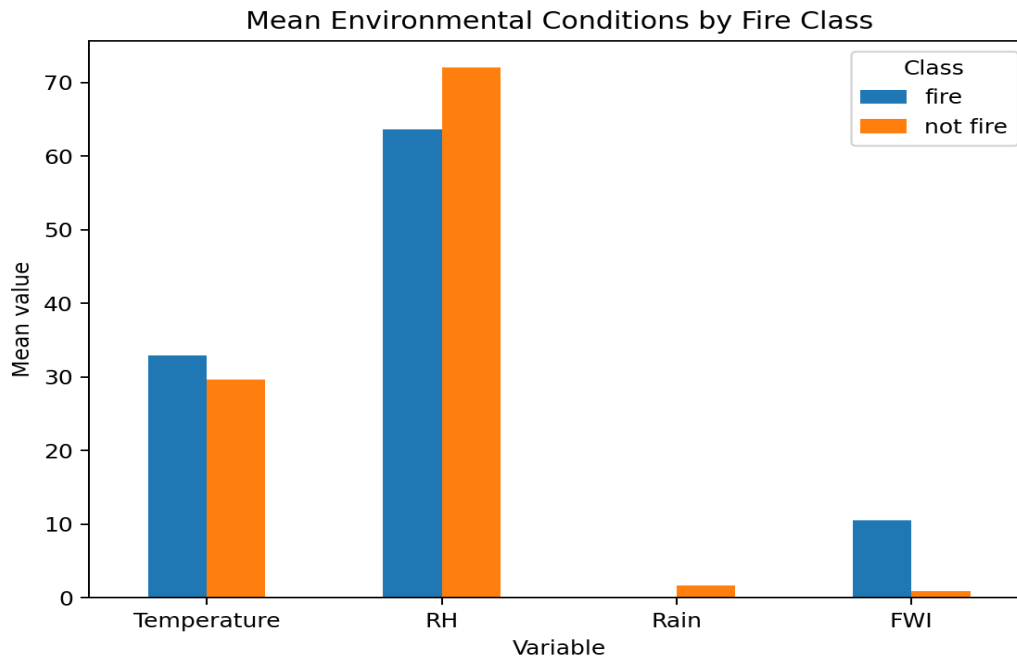
5. Statistical Analysis

Exploratory Pearson correlations and Welch two-sample t-tests were used to compare fire and not-fire observations. The strongest absolute target associations were observed for ISI, FWI, FPMC, DMC and BUI.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.960	1.000	0.917	0.957	0.994
Random Forest	0.960	1.000	0.917	0.957	0.987

6. Interpretation

On the fixed test split, both models achieved 96.0% accuracy. Both had 100% precision and 91.7% recall, with F1-score 95.7%. There were 13 true negatives, 11 true positives, 0 false positives and 1 false negative. These results are specific to the sample and split.



7. Conclusion and Limitations

Weather and Fire Weather Index variables contain useful information for distinguishing fire and not-fire observations in this dataset. The project is an educational prototype. The working data covers one region and a limited 2012 period, so the model should not be treated as an operational fire-warning system or generalized to India without further validation.

8. Future Scope

Integrate NASA FIRMS satellite detections, multi-year weather and air-quality data, India-specific geographic information, time-aware validation, calibration and a geospatial dashboard.

References

UCI Machine Learning Repository — Algerian Forest Fires Dataset:
<https://archive.ics.uci.edu/dataset/547/algerian+forest+fire+dataset>